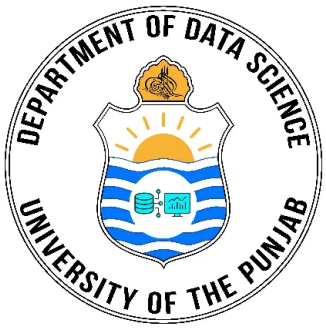
Final Year Project Proposal

RESORA - AI Research Assistant & Knowledge Copilot



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Executive Summary

RESORA is a free, AI-powered research assistant and knowledge copilot designed to improve the way academic literature is discovered, understood, compared, and written. Instead of manually searching papers across many platforms, students and researchers can enter a topic, receive relevant academic papers, read simple summaries, compare related work, identify research gaps, and generate literature review drafts in one connected workflow.

The project improves the previous ReSearch Flow system. ReSearch Flow provided a useful foundation with academic paper search, basic AI summaries, paper chat, citation graphs, user accounts, and export options. However, it had several important limitations. It could not properly understand scanned PDFs, figures, tables, or equations. It worked only in English, had limited paper comparison, did not show confidence in AI answers, and depended on paid tools.

RESORA addresses these limitations using free and open-source technologies. It will support English and Urdu explanations, OCR for scanned papers, figure and table understanding, equation explanation, multi-level summarization, research gap detection, confidence scoring, and literature review generation with citations. The goal is to make academic research easier, faster, and more accessible for students, faculty, and early-stage researchers.

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Chapter 1

INTRODUCTION

1.1 Background

The academic research process is often complex, repetitive, and fragmented. Students and researchers need to explore different online databases, collect relevant papers, understand technical methods, compare related work, and organize citations before they can begin writing a strong literature review. This process becomes even more difficult because new research papers are published every day and academic information is spread across many platforms.

Tools such as Google Scholar, Semantic Scholar, ResearchGate, Mendeley, and Zotero help with individual parts of the workflow. Some tools are good for searching papers, some are useful for managing references, and some can help with writing or grammar. However, most tools do not provide a complete experience from discovery to understanding, comparison, and writing.

With recent progress in Artificial Intelligence, Natural Language Processing, document understanding, and retrieval-based systems, there is a strong opportunity to make research workflows more intelligent. RESORA uses these technologies to help users discover, analyze, summarize, compare, and write about academic literature in a simpler and more reliable way.

1.2 Problem Statement

Researchers, especially undergraduate students, FYP students, and early-career academics, spend excessive time manually finding relevant papers, understanding difficult academic language, comparing different methods, and preparing summaries or literature reviews. Current tools either focus on paper search, reference management, grammar improvement, or general AI assistance. They do not provide one complete free system that supports the full research workflow.

This fragmented process creates several problems. Students may collect too many papers but still fail to understand which ones are important. They may miss research gaps because comparison is done manually. Many papers contain figures, tables, equations, and scanned pages that simple text-based tools cannot read. Pakistani students may also need simple Urdu

explanations, but most research tools only work in English.

RESORA is proposed to solve these problems by combining paper search, AI summarization, paper chat, Urdu support, OCR, multimodal document understanding, comparison, gap detection, and literature review generation in one platform.

1.3 Project Goals and Objectives

The main goal of RESORA is to develop an AI-powered platform that simplifies academic research through automation, intelligent understanding, comparison, and writing support. The system should reduce manual effort while still keeping answers traceable to real paper content.

- Build a multi-source paper search system using free academic APIs.
- Display paper metadata, authorship details, publication information, and citation-related data.
- Generate short, structured, and detailed summaries for research papers.
- Allow users to ask questions about papers and receive source-based answers.
- Support English and Urdu explanations for better local accessibility.
- Extract text from normal PDFs and scanned PDFs using OCR.
- Understand important figures, tables, and equations in academic papers.
- Compare multiple papers by methods, datasets, results, limitations, and future work.
- Detect research gaps and support them with evidence from related papers.
- Generate literature review drafts with citations and export options.

1.4 Scope

RESORA will focus on academic domains such as Data Science, Computer Science, Artificial Intelligence, Machine Learning, and related areas. The system will use free academic APIs such as Semantic Scholar, arXiv, OpenAlex, and Crossref to collect paper metadata and related information. Users will also be able to upload PDF papers for summarization and question answering.

The AI tasks in scope include summarization, question answering, keyword extraction, paper comparison, research gap detection, and literature review generation. The document understanding tasks include normal PDF reading, OCR for scanned PDFs, and basic understanding of figures, tables, and equations.

The current scope does not include paid academic databases, plagiarism checking, journal submission, or replacing expert academic supervision. RESORA is intended to support research work, not to replace the judgment of students, researchers, or supervisors.

1.5 High-level System Components

1.5.1 Web Application

- Topic input page for searching research papers from free academic sources.
- Paper listing page with title, authors, year, source, citation information, and quick summary.
- Paper detail page with structured summary, important methods, limitations, and references.
- Paper chat interface where users can ask questions about uploaded or searched papers.
- Comparison page for viewing multiple papers side by side.
- Literature review generation and export page.

1.5.2 Backend Services

- Academic API connectors for collecting paper data from free sources.
- PDF processing service for text extraction, layout reading, and OCR.
- AI service for summarization, question answering, comparison, and writing support.
- Vector search service for retrieving relevant paper passages before generating answers.
- Database service for user accounts, saved papers, search history, and chat history.

1.6 Optional Functional Units

Some features may be added if time allows after the core system is completed. These include voice-based paper chat, collaborative reading lists, shared literature review editing, mobile-friendly offline reading, and reference manager integration with tools such as Zotero or Mendeley. These features are not required for the first version, but they show the future direction of the project.

1.7 Application Architecture

RESORA will follow a modular web application architecture. The frontend will provide a clean user interface for search, reading, chat, comparison, and export. The backend will handle API requests, user management, PDF processing, and communication with AI services. The AI layer will generate summaries, answers, comparisons, and review drafts. The vector layer will store embeddings and retrieve the most relevant paper passages for source-based responses.

- Frontend Layer: React.js and Tailwind CSS for a responsive web interface.
- Backend Layer: FastAPI in Python for application logic and API endpoints.
- Document Layer: PyMuPDF and Tesseract OCR for PDF and scanned document processing.
- AI Layer: free LLM options such as Groq/Llama for summarization and reasoning.
- Vector Layer: Sentence-Transformers and FAISS for semantic search.
- Data Layer: PostgreSQL or SQLite for users, papers, and history.

1.8 System Limitations and Constraints

RESORA will mainly depend on open-access papers and free academic APIs. Some papers may have incomplete metadata, limited full-text access, or poor-quality scanned pages. OCR accuracy may vary for old documents, blurred scans, or complex layouts. AI-generated answers may also need human review, especially when the content is highly technical.

The project must be completed within the Final Year Project timeline by a four-member undergraduate team. Because of this, the first version will focus on the most useful and practical features. Optional features will be kept for future work if the core system is completed successfully.

1.9 Tools and Technologies

Tool/Technology	**Description and Purpose**
React.js + Tailwind CSS	Frontend interface for search, reading, chat, comparison, and export.
FastAPI	Backend API framework for connecting the frontend, database, AI, and academic APIs.
Semantic Scholar, arXiv, OpenAlex, Crossref	Free academic sources for paper search and metadata.
Groq / Llama / Mistral	AI summarization, chat, comparison, and reasoning through free options.
Sentence-Transformers	Converts paper text into embeddings for semantic search.
FAISS	Stores and searches vectors locally for retrieval-based answers.
PyMuPDF	Extracts text and layout information from PDF files.
Tesseract OCR	Reads scanned or image-based PDF documents.
PostgreSQL / SQLite	Stores users, saved papers, paper metadata, and chat history.

Docker	Helps package and deploy the system easily.

Chapter 2

Related Literature Review

2.1 Related Work

Several academic and commercial tools already support research work, but most of them focus on one specific part of the workflow. Google Scholar is widely used for discovering research papers, but it does not explain paper content, compare papers, or generate literature reviews. Semantic Scholar provides useful metadata, recommendations, and citation information, but it does not provide a complete writing and paper understanding workflow.

Mendeley and Zotero are strong tools for reference management. They help users store papers, manage citations, and organize bibliographies, but they do not provide deep AI-based understanding of paper content. Connected Papers is useful for exploring relationships between papers, but it does not fully summarize papers, explain figures, compare methods, or write literature review drafts.

General AI tools such as ChatGPT can explain topics and help with writing, but they may not always be grounded in real paper content unless users provide the documents. Some AI research tools provide paper summaries, but they may be limited by paid features, lack Urdu support, or fail to process scanned PDFs and visual content.

2.2 Gap Analysis

No single free tool provides all key research support features in one place. A complete student-friendly system should search papers, read uploaded PDFs, explain content simply, support Urdu, process scanned documents, understand figures and tables, compare multiple papers, detect research gaps, and generate literature review drafts with citations.

RESORA fills this gap by combining research discovery, document understanding, AI-based comparison, and writing support into one accessible platform. It also improves the previous ReSearch Flow idea by focusing on free tools, better document understanding, Urdu accessibility, and more reliable source-based answers.

Chapter 3

Use Case Scenarios

3.1 Overview

RESORA is designed to support different academic users in realistic research situations. A student working on a Final Year Project can enter a topic such as "AI in agriculture" and receive relevant papers with simple summaries, important methods, and possible research gaps. This reduces the time spent on initial paper collection and helps the student understand the field faster.

An FYP or thesis student can upload collected PDFs and ask questions about methodology, results, limitations, or future work. If a paper is difficult, the student can request a simpler English explanation or an Urdu explanation. If the paper is scanned, OCR will help extract the text so that the system can still process it.

A supervisor can use RESORA to review whether a student's related work is broad enough. By checking paper comparison, gap detection, and summary outputs, the supervisor can quickly identify weak areas in the student's literature review. Research assistants and study groups can also use the system to prepare shared notes, comparison tables, and early literature review drafts.

- Student Researcher: searches a topic and receives papers, summaries, and research directions.
- FYP Student: uploads PDFs and asks questions in English or Urdu.
- Thesis Supervisor: reviews related work, gaps, and paper coverage more efficiently.
- Postgraduate Scholar: compares methods, datasets, limitations, and future work across papers.
- Study Group: prepares shared notes and literature review material for seminars or projects.

Chapter 4

Target Users and Stakeholders

4.1 Audience Definition

RESORA is mainly designed for students and researchers who regularly work with academic papers. Undergraduate and Master's students can use it for FYPs, assignments, thesis work, and literature review preparation. These users often need simple explanations because research papers can be difficult to understand at the start of a project.

PhD researchers and academics can use RESORA to track related work, compare new studies, and prepare early drafts of literature review sections. Faculty supervisors can use the system to guide students and quickly check whether important papers, methods, and limitations have been considered. Research assistants and academic groups can use it to organize paper summaries and build shared understanding.

- Undergraduate and Master's students working on FYPs, assignments, and thesis projects.
- PhD researchers and academics who need to compare related work.
- Faculty supervisors who guide student research.
- Research assistants and academic groups preparing literature reviews.
- Industry R&D teams that need to understand new techniques before applying them.

Chapter 5

Anticipated Challenges and Risk Mitigation

5.1 Foreseeable Risks

Developing RESORA includes both technical and planning challenges. One major challenge is limited access to full papers because many publications are behind paywalls. To manage this, the system will prioritize open-access papers and use abstracts or metadata when full text is not available.

Another challenge is AI reliability. AI systems can sometimes produce incomplete or inaccurate answers. RESORA will reduce this risk by using retrieval-based answering, showing source references, and providing confidence scores where possible. OCR quality may also vary for old or low-quality scanned papers, so preprocessing and user review will be important.

- Limited paper access: use open-access sources and metadata when full text is unavailable.
- API limits: use multiple free APIs and cache repeated search results.
- AI inaccuracy: use source-based answers, confidence scores, and clear user review.
- OCR quality issues: apply preprocessing and allow manual checking of extracted text.
- Large paper comparison: use batching and optimized vector search.
- Time limits: complete core features first and keep optional features for later.

Chapter 6

Expected Learning Outcomes

6.1 Academic and Technical Skills

This Final Year Project will help the team gain practical experience in applied data science, AI, and full-stack software development. The project combines real-world data collection, document processing, natural language processing, vector search, interface design, and deployment.

Team members will learn how to connect academic APIs, extract and process PDF content, use OCR, create embeddings, build retrieval-based AI systems, and design a usable web application. They will also learn how to test AI outputs, document system design, manage tasks, and work collaboratively through an academic software project.

- Academic API integration and structured data collection.
- PDF processing, OCR, and document understanding.
- Natural Language Processing for summaries, question answering, and review generation.
- Vector search and retrieval-based AI systems using embeddings and FAISS.
- Frontend and backend development using React, Tailwind CSS, Python, and FastAPI.
- Database design for users, papers, saved work, and chat history.
- Testing, documentation, teamwork, and deployment.

Chapter 7

Future Enhancements

7.1 Scalability and Vision

After the current version is completed, RESORA can be extended into a stronger research assistant with more collaboration, personalization, and integration features. Future versions may include shared literature review editing, team reading lists, and supervisor feedback tools. These features would make the system useful not only for individual students but also for research groups.

The system can also be improved with journal and conference recommendations, deeper citation network visualization, plagiarism support, voice-based paper chat, and integration with reference managers such as Zotero and Mendeley. Over time, RESORA can become a complete research companion for students, teachers, and researchers.

- Team collaboration and shared literature reviews.
- Integration with Zotero or Mendeley.
- Journal or conference recommendations.
- Voice-based paper chat.
- Research trend prediction.
- Better citation network visualization.